Highlights

Paper 1: Soil Hydraulic Memory and the Recoverability of Evaporation Savings

- Hydraulic memory: a dimensionless diagnostic of how soil pores route saved water
- Memory fate spans 0.02 to 0.85 across ten soils and follows pore-size parameter n
- One-parameter rate-competition law predicts drainage of unseen soils (RMSE 0.058)
- The recoverable share of an evaporation saving is soil- and climate-specific
- Two soil properties give planners a discount for evaporation-saving claims

Paper 2: Auditing Water-Balance Closure in a Soil-Plant Model with Interpretable Machine Learning

- A blind pipeline recovers the water-balance law against a permutation null
- Closure audit exposes an unreported evapotranspiration term of about 30% of ET
- Hand-checkable three-term ET law matches gradient boosting (LOEO R^2 0.898)
- Positive control, null model and regime-wise validation define a discovery protocol
- Sparse laws come with a stated domain of validity, not as universal constants

Paper 3: Modeling Soil–Water–Atmosphere Interactions for Irrigation Scheduling and Design

- Every output of an offline irrigation tool is tied to a numbered equation
- 134 equations carry provenance labels: 38 literature, 18 data-driven, 78 engineered
- Meta-model of 10,190 Daisy runs ranks held-out scenarios (Spearman 0.64-0.97)
- Surface, drip and sprinkler design reproduce FAO-45 and textbook examples within 3%
- One command re-runs every check; 10^5 scenario sweeps take about 2 s